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Transport NOx Emission Reduction Barriers: A DEMATEL-Based Causal Investigation

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Abstract-Urban air quality has worsened in ways that decades of emission legislation have failed to fully reverse. Nitrogen oxides from transport operations sit at the core of this problem — not because technical solutions are absent, but because the route from regulatory intent to real atmospheric outcomes is blocked by a network of interconnected barriers that resist piecemeal policy fixes. This study examines fifteen transport NOx reduction barriers through the Decision-Making Trial and Evaluation Laboratory approach, drawing on structured pairwise influence assessments from four professionals spanning vehicle engineering, regulatory compliance, transport policy, and fleet operations. The analysis establishes a clear causal hierarchy. Lobbying and industry resistance dominates the system with a net causal score of +3.700 — a value so far above everything else in the dataset that it occupies a structural category of its own. Driver behaviour and hardware degradation follow as meaningful secondary causes at +1.000 each, while fuel quality contributes causal pressure at +0.450. Maritime regulatory gaps hold the system's highest prominence score at 5.994 while belonging firmly to the effect group, alongside policy and incentive deficiencies at 5.500. The central finding is one that conventional transport emission policy has been reluctant to absorb: the most visible and frequently discussed barriers are largely downstream products of governance failures operating upstream, and resolving those upstream failures is a precondition for any downstream improvement to hold.

Keywords: transport NOx barriers; DEMATEL; emission governance; cause-effect analysis; air quality management; lobbying resistance; driver behavior; barrier ranking

I. Introduction

Something has gone persistently wrong in the translation of transport emission policy into real atmospheric outcomes. Regulatory frameworks governing nitrogen oxide emissions from road vehicles, maritime operations, and freight transport have grown considerably more elaborate over the years — and yet the gap between what those standards imply and what actually appears in ambient air measurements has remained frustratingly wide across most major transport corridors. Understanding why requires looking past the regulatory text and asking what structural conditions are consistently getting in the way.

NOx from transport is not a peripheral concern. These compounds trigger atmospheric reactions producing ground-level ozone and secondary fine particles, both carrying serious consequences for respiratory function, cardiovascular health, and ecosystem integrity [1, 2]. Port cities and urban freight corridors experience concentrated exposures precisely where population density and atmospheric conditions combine to maximise harm [3]. Successive regulatory frameworks — Euro emission standards, MARPOL Annex VI NOx limits, real-world driving emission requirements — have been introduced with genuine environmental ambition, and vehicle technology has genuinely improved in response. The problem is that real-world fleet emission performance has consistently failed to track certified improvements at the rate policy trajectories assumed [4].

A structural implementation gap of this persistence demands structural analysis. Treating each obstacle separately — a testing protocol that needs revision here, an enforcement capacity that needs strengthening there — consistently misses the connections between obstacles that determine how the system actually behaves. An enforcement authority operating under a regulatory framework shaped by heavy industry influence will produce compliance results bounded by that framework's design ceiling, regardless of how vigorously individual violators are pursued. A fleet renewal programme will disappoint if driver behaviour patterns stress replacement vehicles' hardware at rates that push real-world performance below certified levels within a few operating years.

RQ1: What are the critical challenges that need to be considered in the selection of HR practices in start-ups?

RQ2: What are the cause-and-effect relationships among the identified challenges?

RQ3: What measures should be taken by HR persons to mitigate or eliminate these critical challenges?

This study takes the causal architecture of transport NOx barriers as its central subject. Through DEMATEL — a structured influence-mapping method producing quantitative prominence and causal-direction scores from expert-assessed pairwise influence data — fifteen identified barriers are analysed for their structural roles within the system [5, 6]. Which barriers are driving the others? Which are being driven? Where should policy attention go first to produce the widest systemic improvement? These are the questions the analysis is designed to answer, and the findings carry direct implications for how emission policy resources and regulatory reform efforts should be sequenced.

II. Literature Review

The Gap Between Certified and Real-World Performance

The relationship between vehicle type-approval emission figures and what roadside sensors actually measure has been studied extensively enough that the divergence between them can no longer be treated as a measurement anomaly — it is a fundamental structural feature of how transport emission regulation has been designed. Research using remote sensing campaigns at urban locations found that diesel passenger cars in everyday service produced NOx outputs several times their certified limits, with the divergence growing wider as successive standards nominally tightened those limits [7]. The paradox of stricter standards producing larger real-world gaps rather than smaller ones pointed toward something systematic in the regulatory architecture rather than random non-compliance.

What produces this divergence is a combination of reinforcing factors. Test procedures have historically been designed around conditions considerably more benign than those vehicles routinely encounter — controlled temperatures, prescribed acceleration sequences, fixed load patterns. Engine management software is calibrated against those conditions, producing emission behaviour during testing that does not reflect how the same software responds under hard acceleration in cold weather, heavy freight loads on motorway gradients, or the stop-start demands of urban distribution [8]. Beyond testing design, fleet age compounds the picture: the air quality benefits of tighter standards only materialise when new compliant vehicles actually replace older high-emitting ones, and modelling of European fleet renewal rates found this substitution process consistently delayed air quality improvements by years beyond standard implementation dates [9].

Economic and Regulatory Barriers

The economics of cleaner technology adoption look

compelling at the aggregate societal level, where health and environmental costs of NOx exposure are counted alongside private equipment costs. They look far less compelling from the perspective of an individual fleet operator with constrained capital, competitors potentially not facing the same investment pressure, and no direct financial exposure to the external costs their vehicles impose. This private-social incentive gap sits at the root of many economic NOx barriers and cannot be closed through technology improvement alone [10].

Regulatory design creates its own structural vulnerabilities. Mock and colleagues assessed type-approval processes across major vehicle markets and found technical assumptions that systematically favoured optimistic emission estimates, with the gap between approved and real-world performance widening as test conditions became more prescriptive and therefore more amenable to targeted engine calibration [11]. Defeat devices — software mechanisms detecting official test conditions and activating emission controls suppressed during normal operation — were found deployed across multiple manufacturers and vehicle categories following investigations that established the practice had a considerably longer history than initial disclosures suggested [12]. Enforcement faces compound structural challenges: vehicles generating the highest emissions are distributed across millions of operators, cross jurisdictional boundaries, and operate in conditions making point-in-time compliance checks difficult to interpret meaningfully [13].

Political Economy of Standard-Setting

The processes through which transport emission standards are established are not technically insulated exercises — they are political and administrative processes involving sustained engagement from regulated industries, with substantial asymmetries in expertise, organisational capacity, and financial resources between regulatory bodies and those they regulate [14]. Oreskes and Conway documented how industry-funded strategies have introduced manufactured uncertainty into emission policy debates — commissioning alternative technical studies, placing sceptical commentary through affiliated organisations, mobilising economic impact arguments at legislative hearings — proving effective at extending the interval between a problem being identified and a regulatory response being enacted [15].

Cashore and Howlett analysed regulatory change processes in industries with strong organised interests and found that outcomes systematically reflected the distribution of political capacity among participants rather than solely the technical merits of options under consideration [16]. In the transport sector, manufacturer associations have participated in test procedure development in ways introducing flexibility that accommodated non-compliant engine calibration strategies; fuel industry bodies have engaged in quality standard-setting in ways preserving supply chain cost structures at the expense of sulphur reductions that emission control technology most needs; shipping industry organisations have engaged international maritime regulatory processes in ways slowing NOx control area expansion [17].

Behavioural and Technical Dimensions

Driver behaviour as a determinant of real-world emission output receives less policy attention than its significance warrants. Research instrumenting vehicles to capture real-time emission data alongside driving behaviour metrics found NOx output differences between driving styles exceeding a factor of three under comparable route conditions, with aggressive-driving scenarios pushing even modern vehicles well above certified limits [18]. The aggregate emission performance of any vehicle fleet is therefore determined substantially by the distribution of driving behaviours across operators — a distribution shaped by training practices, economic incentive structures, and operational pressures that regulatory frameworks have not systematically addressed.

Hardware degradation over operational life is closely connected to this through the thermal and mechanical stress that aggressive operation accelerates. SCR catalysts, EGR valves, and diesel particulate filters all decline in performance over time, and that decline proceeds faster under real-world conditions than controlled laboratory durability testing predicts [19, 20]. Fuel quality functions as a technical prerequisite for emission control rather than a parallel concern — SCR systems require diesel sulphur content below fifty parts per million for sustained effective catalyst operation, and in supply chains where quality assurance is inconsistent, vehicles that were certified as compliant progressively drift into non-compliance through chemistry rather than through any deliberate operational choice [21].

Social Acceptance and Analytical Frameworks

Public acceptance has sometimes proven the binding constraint on implementation. Research investigating how people evaluate transport emission policies found that perceived distributional fairness — whether costs fall equitably across different groups — was more powerful than the environmental evidence for a measure's effectiveness in determining whether it attracted sustainable political support [22]. Policies appearing to penalise economic vulnerability generated resistance that information campaigns could not overcome because the objection was not to the emission reduction objective but to the instrument's distributional consequences.

Structured decision methods have been applied to emission barrier systems with increasing sophistication. Research in supply chain and logistics contexts found that methods capturing interdependence between barriers consistently identified governance and institutional factors as primary causal drivers, with technical and operational barriers clustering on the effect side as downstream consequences [23, 24]. Applications in transport emission contexts confirmed that the bottleneck in emission governance is rarely the availability of technical solutions but the economic, regulatory, and political conditions determining whether those solutions are adopted — findings that provide the analytical foundation for the present study's use of DEMATEL [25].

III. NOx Emission Reduction Barriers

Fifteen barriers were identified through systematic literature review and refined through consultation with four domain experts — a transport engineer with seventeen years of aftertreatment system experience, a regulatory policy analyst with a decade in compliance programme design, an academic transport policy researcher, and a commercial fleet operations manager. Table 1 presents the finalised barrier set.

IV. Methodology

Rationale for DEMATEL

The analytical challenge here is not simply to rank fifteen barriers by importance but to understand how they relate to one another causally. Many barriers that dominate policy discussions — public resistance, hardware degradation, defeat devices — may be downstream consequences of

deeper governance failures rather than independent problems requiring standalone solutions. A method that treats all barriers as independent cannot reveal this structure and will consistently direct policy toward symptoms rather than sources.

DEMATEL addresses this directly. By requiring experts to assess how strongly each barrier influences every other barrier, and by computing both direct and indirect influence pathways through matrix operations, it produces a quantitatively grounded causal architecture distinguishing barriers that are driving the system from those being driven by it [5, 6]. For transport NOx barrier analysis — where the most consequential connections are often not the most immediately obvious — this causal directionality capability is analytically essential.

Data Collection

Each expert received a structured 15×15 questionnaire with written barrier definitions and was asked to rate direct influence between every ordered barrier pair using the five-point scale in Table 2. Diagonal elements were pre-set to zero.

Computational Procedure

Step 1 — Expert Matrices: Each expert's responses were arranged into a 15×15 direct-influence matrix.

Step 2 — Average Matrix: Four matrices were averaged element-wise:

$$a_{ij} = \frac{1}{4}\sum_{k=1}^{4} x_{ij}^{k} \qquad (1)$$

aij=41*k*=1∑4*xijk*(1)

Step 3 — Normalisation:

$$m = \min\left[\frac{1}{\max_j \sum_i a_{ij}}, \frac{1}{\max_i \sum_j a_{ij}}\right] \qquad (2)$$

m=min[max*j*∑*iaij*1, max*i*∑*jaij*1](2)

$$\mathbf{N} = m \cdot \mathbf{A} \qquad (3)$$

N=*m*·**A**(3)

Step 4 — Total Relation Matrix integrating all direct and indirect pathways:

$$\mathbf{T} = \mathbf{N}(I - \mathbf{N})^{-1} \qquad (4)$$

T=**N**(*I*−**N**)−1(4)

Step 5 — Row and Column Sums capturing outward and inward influence:

$$R_i = \sum_{j=1}^{15} t_{ij}\,, C_i = \sum_{i=1}^{15} t_{ij} \qquad (5, 6)$$

Ri=*j*=1∑15*tij*,*Ci*=*i*=1∑15*tij*(5, 6)

Step 6 — Prominence and Net Effect:

$$\text{Prominence}_i = R_i + C_i \qquad (7)$$

Prominence*i*=*Ri*+*Ci*(7)

$$\text{Net Effect}_i = R_i - C_i \qquad (8)$$

Net Effect*i*=*Ri*−*Ci*(8)

Positive net effect values identify **Cause barriers**; negative values identify **Effect barriers**.

V. Results and Discussion

Prominence Rankings

B10 (Maritime Gaps) holds the highest overall prominence at 5.994, which immediately raises an analytical point worth examining: the most systemically connected barrier in the network belongs to the effect group. Prominence and causal direction measure different things — prominence captures total mutual influence density regardless of directionality, while net effect captures which direction dominates. A barrier can be densely connected across the system while being primarily on the receiving end, and that is precisely what B10's profile describes.

B14 (Driver Behaviour) at 5.800 and B7 (Fuel Quality) at 5.790 complete the top three — both in the cause group. Having two of the three most prominently connected barriers driving the system rather than responding to it is analytically significant. At the opposite end, B5 (Additive/Infrastructure) at 2.497 and B1 (RDE Gap) at 3.294 are the most peripherally connected barriers — real practical constraints, but not the nodes through which the most consequential system-level influence flows.

Cause Group Analysis

B15 — Lobbying and Industry Resistance

A row sum of 4.650 against a column sum of just 0.950 produces a net-effect value of +3.700 sitting in a category of its own. The nearest cause-group competitors — B14 and B13 both at +1.000 — are less than a third of B15's causal magnitude. This is not a marginal difference but a structural gulf reflecting the qualitatively distinct character of industry lobbying as a systemic force.

B15 achieves this pervasive reach not through overwhelming influence on any single barrier but through simultaneously shaping conditions across virtually every domain: regulatory design (B1, B6), technical standard-setting (B7), compliance monitoring (B9), maritime governance (B10), policy architecture (B12), and incentive structures (B3, B4). When industry lobbying successfully shapes any one of these domains, it creates a more permissive environment for non-compliance throughout the rest. The policy implication is direct: downstream interventions — better testing technology, stricter enforcement, improved incentive schemes — all produce results bounded by the regulatory ceiling that the standard-setting process sets. If that ceiling has been shaped by interests invested in keeping it manageable, more vigorous action beneath that ceiling will produce correspondingly capped results.

B14 — Driver Behavior

At prominence 5.800 and net effect +1.000, driver behavior combines the second-highest systemic centrality with meaningful causal directional influence. Its most consequential causal pathways connect to B13 (Hardware Degradation) and B8 (Thermal Management). Hard acceleration and sustained high-speed operation generate thermal and mechanical stress on SCR catalysts, EGR valves, and particulate filter assemblies that drives performance degradation faster than durability approval testing anticipates. Cold-start patterns prevent aftertreatment systems reaching the operating temperatures their chemistry requires. These are not exceptional conditions — they describe daily operating realities for commercial vehicles across urban distribution and motorway freight routes.

A less obvious but consequential B14→B9 pathway also operates: driving patterns routinely placing vehicles outside the conditions emission controls are calibrated for create a grey zone where the boundary between legitimate engine management and prohibited defeat strategy becomes practically difficult to police. This cross-domain connection — behavioral barrier providing indirect cover for regulatory compliance failure — is exactly the kind of structural relationship that causal mapping surfaces and simpler ranking methods cannot.

B7 — Fuel Quality
At prominence 5.790 and net effect +0.450, fuel quality holds the third-highest prominence while maintaining cause-group positioning. Its causal influence flows toward B4 (Zero-Emission Technology adoption), B5 (Additive/Infrastructure), and B13 (Hardware Degradation). The fuel quality-hardware degradation relationship is a direct chemical dependency: SCR systems operating above fifty parts per million Sulphur threshold face catalyst poisoning accelerating performance loss beyond what type-approval durability testing predicts. In supply chains where quality assurance is inconsistent or where adulteration is commercially widespread, vehicles certified as compliant progressively drift into non-compliance through chemistry rather than deliberate choice. Binding minimum fuel quality specifications enforced through systematic supply chain surveillance — rather than aspirational guidance — create the technical preconditions that multiple downstream barriers depend on.

B13 — Hardware Degradation
Hardware degradation at prominence 4.800 and net effect +1.000 occupies a cause-group position that challenges its conventional framing as a purely operational maintenance challenge. Its outward causal influence connects to B2 (Fleet Turnover) through the economic mechanic of persistent hardware failure: operators facing repeated SCR catalyst replacements and EGR valve failures on otherwise serviceable vehicles encounter a capital allocation calculation where premature retirement may cost less than continued compliance maintenance. Fleet turnover is therefore partly driven by hardware reliability as well as by age and mileage. The B13→B9 pathway describes a further dynamic: when hardware failure rates are high and repair costs significant relative to operating margins, the financial incentive to disable aftertreatment systems creates field-level defeat conditions independently of any manufacturer-programmed strategy.

B6 and B3
Enforcement records net effect +0.846 and ranks thirteenth in prominence. Its causal direction toward B9 and B1 is genuine but its systemic influence is bounded by the regulatory standards it enforces — standards whose ambition is itself shaped by the upstream lobbying dynamics in B15. Retrofit cost at +0.347 and eleventh place exerts modest causal pressure toward B2 and B4: high retrofit investment requirements competing with fleet renewal and zero-emission vehicle purchasing capital slow both pathways to emission improvement simultaneously.

Effect Group Analysis
B10 — Maritime Gaps
At prominence 5.994 and net effect −1.500, maritime regulatory gaps represent simultaneously the most systemically connected and most reactive barrier in the analysis. The persistence of maritime NOx regulatory shortfalls is not primarily a vessel-level technical or monitoring problem — it is a governance problem rooted in fragmented international regulatory authority and organised shipping industry resistance to NOx control area expansion. The B15→B10 causal pathway runs through the same lobbying dynamic operating through IMO standard-setting processes that has moderated maritime emission ambition relative to what atmospheric chemistry and public health require. Port-level monitoring improvements and expanded emission control areas will contribute, but results will remain bounded while B15 and B12 continue operating at their current levels of causal influence.

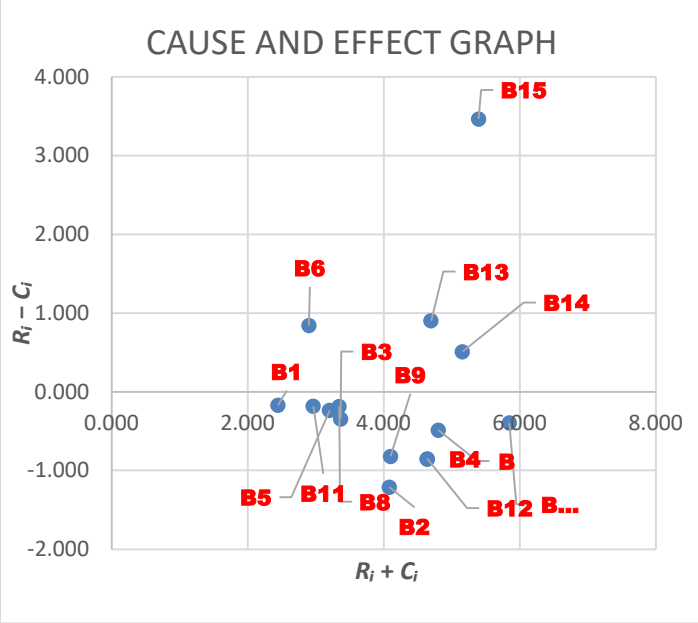
B12 — Policy and Incentives
Policy and incentive deficiencies at prominence 5.500 and net effect −1.394 capture the accumulated institutional shortcomings of transport emission governance as reactive consequence rather than independent failure. The barriers flowing most strongly into B12 include B15 (limiting political feasibility of ambitious incentive programs), B14 (generating operational demands that incentive design must navigate), and B7 (limiting relevance of incentive structures predicated on clean-fuel availability). Better incentive design matters and should be pursued — but expecting it to deliver transformative improvements before B15's influence on the design process is addressed means expecting policy to overcome the conditions of its own creation.

B9 — Defeat Devices
Defeat devices at prominence 4.400 and net effect −1.300 are effect barriers in a finding that reframes how the defeat device problem should be understood. Detection and punishment are necessary responses, but the DEMATEL classification points toward the more fundamental issue: defeat devices persist because the conditions making them technically feasible and financially rational have not been eliminated. Those conditions — test procedures exploitable through targeted calibration (B1), enforcement capacity insufficient to detect software-based strategies (B6), and standard-setting shaped by actors invested in preserving the laboratory-road gap (B15) — are embedded in the causal barriers upstream. Closing the defeat device problem means closing the conditions sustaining it.

Remaining Effect Barriers
Zero-emission technology adoption barriers (B4) at 4.730 and −0.850 reflect investment decisions responding to the policy incentive landscape and infrastructure quality rather than driving them. Fleet turnover (B2) at 4.200 and −1.126 follows the economic logic of operators responding to hardware reliability costs and incentive signals originating upstream. Thermal management challenges (B8) at 3.780 and −1.000 represent the downstream technical consequence of driver behavior and fuel quality constraints combining to stress systems at the edges of their design envelopes. Public resistance (B11) at 3.200 and −1.100 is the social expression of policy design failures and distributional concerns originating in institutional conditions rather than in fundamental opposition to emission reduction goals. RDE gap (B1) at 3.294 and −0.652 and additive/infrastructure constraints (B5) at 2.497 and −1.203 are the most peripheral barriers — real but occupying positions at the edges of the causal network where they are more responsive to changes elsewhere than capable of generating independent systemic change.

Cause-and-Effect Graph
The cause-and-effect graph positions each barrier with $(R_i + C_i)$ on the horizontal axis capturing systemic centrality and $(R_i - C_i)$ on the vertical axis capturing causal direction. Cause barriers appear above the horizontal midline; effect barriers below.
The dominant visual feature is B15's position at

approximately (5.600, +3.700) — isolated far above the rest of the cause group by a margin making its structural separation immediately legible. The nearest cause-group barriers, B14 and B13 both at +1.000, sit at less than a third of B15's vertical elevation. Among effect barriers, B10 at (5.994, −1.500) occupies the rightmost and lowest position, communicating the combination of maximum systemic engagement and strongly reactive causal positioning that characterizes the most consequential downstream barriers. B5 and B1 in the lower-left quadrant reflect the peripheral and reactive character of the system's least connected barriers.



Policy Implications

The causal architecture revealed here supports a specific and practically grounded sequencing logic for transport NOx policy.

Governance reform must come first. B15's causal dominance means every other intervention operates within a ceiling set by the regulatory framework's quality. Mandatory public disclosure of industry submissions to standard-setting processes, independent scientific advisory authority for emission limit baselines, formal conflict-of-interest protocols for regulatory working group participants, and civil society representation mechanisms in consultations would each raise that ceiling — and doing so would simultaneously improve the conditions experienced by multiple downstream effect barriers.

Operational accountability must extend beyond vehicle specification. B14's contribution to hardware degradation and field-level defeat behavior calls for emission frameworks holding operators accountable for operational performance rather than only vehicle purchase decisions. Remote sensing surveillance at scale, operator emission performance registries linked to licensing decisions, and incentive structures rewarding demonstrated operational compliance are mechanisms that engage B14's causal role in ways that training programs alone cannot.

Fuel quality must be treated as foundational infrastructure. B7's causal connection to hardware performance and zero-emission technology adoption argues for binding minimum standards enforced through systematic supply chain surveillance — the technical preconditions that multiple downstream barriers depend on cannot be reliably established through voluntary quality assurance alone.

Hardware accountability should extend through operational life. Addressing B13's causal contribution to fleet economics and field-level defeat incentives requires extended manufacturer warranty obligations for emission-related hardware, third-party durability requirements in type-approval procedures, and accessible maintenance compliance funding for small operators unable to absorb recurring hardware replacement costs.

Effect-side interventions should be sequenced after upstream causes are addressed. Maritime gap reduction, incentive programmer improvement, defeat device detection enhancement, and fleet renewal support all represent worthwhile investments — but each will deliver greater and more durable results when the governance and operational conditions sustaining their persistence have changed. Interventions targeted at effect barriers continuously regenerated by unaddressed causal forces will produce improvements that are real but bounded.

VI. Conclusion

The transport NOx emission challenge has resisted straightforward solutions long enough that the productive question is no longer why individual interventions underperform but what structural features of the barrier system have consistently limited their cumulative effect. This study addressed that question by applying DEMATEL to fifteen identified barriers, producing a quantitative causal architecture from expert influence data that reveals both the prominence ranking of every barrier and — more importantly — which are driving the system and which are being driven by it.

Lobbying and industry resistance (B15) is the primary causal force by a margin that reflects not just direct influence on individual barriers but simultaneous reach across regulatory, technical, economic, and social domains. Its net-effect value of +3.700 against a row sum of 4.650 and column sum of just 0.950 reflects extreme asymmetry between outward influence and inward conditioning that places it structurally apart from every other barrier in the analysis. Driver behavior and hardware degradation follow at +1.000 each, with fuel quality contributing causal pressure at +0.450. These four cause-group barriers collectively shape the environment experienced by nine effect-side barriers representing the most visible manifestations of the NOx reduction challenge.

Maritime gaps (B10) being the highest-prominence barrier at 5.994 despite effect-group classification reframes how maritime NOx governance should be understood — not as a self-contained sectoral problem but as the most visibly prominent downstream consequence of governance failures distributed throughout the system. Policy and incentive deficiencies (B12) at fifth prominence and second-most-severe effect classification similarly reveal institutional shortcomings as reactive products of upstream causal conditions rather than independent failures amenable to standalone correction.

The practical contribution of this study lies in providing structural and quantitative grounding for intervention sequencing decisions typically made on the basis of intuition or political convenience. Knowing that B15 carries +3.700 while B10 carries −1.500 provides an evidence-based argument for pursuing governance reform before maritime monitoring investment — not because maritime monitoring is unimportant but because its effectiveness will be limited until the governance conditions sustaining maritime regulatory gaps are changed.

Future research could extend this framework through longitudinal application tracking how the causal architecture evolves as major regulatory reforms are enacted, comparative cross-national analysis assessing whether lobbying's causal dominance is a universal structural feature, and fuzzy DEMATEL extensions handling expert judgment uncertainty more explicitly. What the present study contributes is a rigorously derived baseline causal map placing governance integrity at the centre of the transport NOx challenge — and providing quantitative foundations for the policy priorities that follow from that positioning.

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